

Bufotalin 自主预算耗尽版 Agentic Blackboard 报告

真实运行：action 由默认 blackboard planner 选择；本地 PDF 只在 scout 选中后进入流程

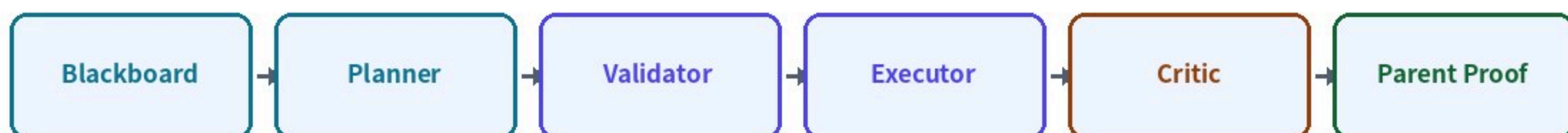
目标 bufotalin

SMILES 重原子 / 环数 32 / 5

最终结论 unresolved (child_solved_parent_unresolved)

停止原因 max_round_budget_exhausted, rounds 8/8

本地 PDF /root/autodl-tmp/AutoPlanner/1-s2.0-S0040402025001668-main.pdf (878443 bytes)



关键边界

本报告展示的是持续探索到轮次预算耗尽，不是提前 stop。即使过程中出现文献候选、类比排序、guided ChemEnzy 或子目标尝试，最终 solved 仍只由 stitched_parent_route_proof.v1 决定。

为什么这一轮不是直接停止

默认 blackboard planner 自主选 action；预算耗尽模式只改变停止策略

1. 实际停止原因：max_round_budget_exhausted。

2. 轮次预算：8/8，budget_exhausted=True。

3. action batch 每轮仍经过 schema、预算和 raw reaction 注入校验；planner 不能直接宣称 solved。

4. 本地 PDF 没有预先塞给视觉工具；只有 `search\_literature` 选择它以后，下一轮才执行 PDF 渲染。

5. 当普通策略会早停时，exhaust 模式先改走未耗尽的替代 action；没有 parent proof 时 final verdict 仍保持 non-solved。

Action 决策轨迹

每个 action 都有 rationale、expected_artifact、success_condition 和执行记录

第 1 轮 - 生成目标侧断键假设 [accepted, useful=True] 理由：target-side bridge tasks are missing
第 1 轮 - 检索目标近端文献 [accepted, useful=True] 理由：blackboard lacks target-proximal literature/source evidence
第 2 轮 - 渲染/索引本地 PDF 结构证据 [accepted, useful=True] 理由：local PDF source is available and must be converted into current-run visual evidence
第 2 轮 - 排序类比假设 [accepted, useful=True] 理由：advisory target-side hypotheses can be ranked before rerun selection
第 3 轮 - 视觉抽取文献结构链 [accepted, useful=True] 理由：source candidates exist but exact rows are missing
第 4 轮 - 视觉抽取文献结构链 [accepted, useful=True] 理由：prior visual extraction left target-relevant source-detail gaps; repair those gaps before compiling exact rows
第 5 轮 - 编译 exact 文献行 [accepted, useful=True] 理由：visual chain needs exact source-detail rows for plugin use
第 6 轮 - 运行 guided ChemEnzy [accepted, useful=True] 理由：bridge tasks and search hints are available for one guided rerun
第 7 轮 - 扩展上游子目标 [accepted, useful=True] 理由：advanced terminal/upstream bridge task exists
第 8 轮 - 拼接父路线证明 [accepted, useful=True] 理由：guided/child/literature artifacts need deterministic parent connectivity proof

文献与 PDF 证据链

agent 请求文献后，用户提供的本地 PDF 才进入抽取流程

Scout source candidates

- doi:10.1016/j.tet.2025.134610: local_pdf=True, recommendations=extract_pdf_literature_structures, extract_visual_literature_chain, compile_exact_literature_rows
- query:bufotalin:polycyclic_cage_core: local_pdf=False, recommendations=extract_pdf_literature_structures, extract_visual_literature_chain, compile_exact_literature_rows
- query:bufotalin:protecting_group_level_transformations: local_pdf=False, recommendations=extract_pdf_literature_structures, extract_visual_literature_chain, compile_exact_literature_rows

PDF render/index summary

- present=True, accepted=True, source=/root/autodl-tmp/AutoPlanner/1-s2.0-S0040402025001668-main.pdf
- counts={'compound_text_snippet_count': 200, 'indexed_image_count': 0, 'rendered_page_count': 4, 'scheme_crop_count': 3}
- reasons=无

Visual chain extraction

- present=True, accepted=False, status=completed
- candidate_step_count=3
- reasons=visual_literature_chain_extraction_gaps

真实工具执行摘要

这里列出的都是 controller 记录的 tool_calls，不是报告脚本伪造结果

Tool calls

- extract_pdf_literature_structures: status=accepted, accepted=True, elapsed=0.812s, reasons=无
- extract_visual_literature_chain: status=rejected, accepted=False, elapsed=321.713s, reasons=visual_literature_chain_extraction_gaps, visual_literature_chain_has_no_steps, visual_literature_chain_missing_expected_labels
- extract_visual_literature_chain: status=rejected, accepted=False, elapsed=448.383s, reasons=visual_literature_chain_extraction_gaps
- compile_source_detail_chain_route: status=accepted, accepted=True, elapsed=0.021s, reasons=无
- run_guided_chemenzy_rerun: status=accepted, accepted=True, elapsed=227.942s, reasons=高级同骨架终端被拒绝, 出现无法解释的大重原子跳跃, 文献模板插件未被后端调用, no_verifier_accepted_stock_closed_route
- run_route_expansion_subgoal_search: status=accepted, accepted=True, elapsed=235.588s, reasons=没有子目标被 verifier 证明闭合
- stitch_literature_chain_with_subgoal_route: status=rejected, accepted=False, elapsed=0.022s, reasons=literature_terminal_subgoal_target_mismatch, subgoal_route_not_solved, subgoal_verifier_not_accepted

Artifact result snapshot

- compiled_exact: present=True, accepted=True, reasons=[]
- guided_chemenzy: present=True, accepted=False, route_status=fake_closed_rejected
- route_expansion: present=True, accepted=False, subgoals=1
- stitched_route: present=True, accepted=False, reasons=['literature_terminal_subgoal_target_mismatch', 'subgoal_route_not_solved', 'subgoal_verifier_not_accepted']

与之前 bufotalin 成功版对比

之前的 solved 来自 strict visual chain + compound 11 子目标 + stitch proof

关键事实

- 旧成功 clean verdict: verdict=solved, solved=True
- 旧成功 strict visual chain: accepted=True, steps=15, terminal=strict visual terminal 11
- 旧成功 source-detail exact: accepted=True, one_step_rows=15, terminal_reached=True
- 旧成功 stitch: accepted=True, stock_audit=True, status=solved
- 本轮自主 run: final=unresolved, exact_rows=3, proof_status=child_solved_parent_unresolved

失败反思

- 旧成功不是 ChemEnzy 原生直接闭合 bufotalin，而是 stock -> compound 11 -> bufotalin 的拼接半合成证明。
- 本轮自主 run 没有重新得到 accepted 的 15 步 strict visual/source-detail chain；compile exact rows 因此没有可用行。
- 没有 exact 文献段连到父路线时，guided ChemEnzy 或 child target 的候选只能作为探索反馈，不能作为 parent solved proof。

Blackboard 与最终门禁

失败、类比和 exact evidence 都只能进入状态；solved 需要 parent proof

Bridge tasks

- target_proximal_bridge: polycyclic_cage_core -> target-proximal cage intermediate with matched ring system
- target_proximal_bridge: protecting_group_level_transformations -> same-core alcohol/phenol protection-state analog
- target_proximal_bridge: analogue_diene_iminium_scaffold_inspiration -> target-proximal cage precursor, not a distant analogue product
- target_proximal_bridge_required: target_side -> Find a target-proximal intermediate that explains the large skeleton change.
- bridge_to_literature_product_required: target_side -> Find a target-side bridge before replaying exact source products.
- upstream_terminal_synthesis: advanced_same_scaffold_terminal -> Find upstream synthesis for rejected same-scaffold terminal.

Selected analogical hypotheses

- target_side_analogue_diene_iminium_scaffold_inspiration score=20 verify=target_proximal_bridge, deterministic_parent_proof
- target_side_analogue_diene_iminium_scaffold_inspiration score=20 verify=target_proximal_bridge, deterministic_parent_proof
- target_side_polycyclic_cage_core_preservation score=20 verify=core_retention_audit, max_unexplained_heavy_atom_jump, route_verifier_acceptance

Parent proof gate

- accepted=False, solved=False, route_status=child_solved_parent_unresolved
- reasons=子目标路线未连到父路线桥接点, exact 文献片段未连到父路线, 父路线 verifier 未接受, 库存审计未通过, unexplained_large_atom_jump
- final=unresolved, solved=False

审计文件与验证

报告输出、run artifacts 和 focused tests

测试：full retry v7 running; focused tests passed: `python -m pytest tests/test_agentic_blackboard_controller.py tests/test_failure_critic.py tests/test_parent_route_proof.py -q`

Blackboard artifact refs

- `r1_generate_disconnection_hypotheses_target_side_disconnection_hypotheses_v1: /root/autodl-tmp/AutoPlanner/results/shared/bufotalin_agentic_blackboard_full_retry_v7_20260609/r1_generate_disconnection_hypotheses_target_side_disconnection_hypotheses_v1.json`
- `r1_search_literature_literature_scout_report_v1: /root/autodl-tmp/AutoPlanner/results/shared/bufotalin_agentic_blackboard_full_retry_v7_20260609/r1_search_literature_literature_scout_report_v1.json`
- `r2_extract_pdf_literature_structures_literature_pdf_structure_evidence_v1: /root/autodl-tmp/AutoPlanner/results/shared/bufotalin_agentic_blackboard_full_retry_v7_20260609/r2_extract_pdf_literature_structures_literature_pdf_structure_evidence_v1.json`
- `r2_rank_analogical_hypotheses_analogical_hypothesis_ranking_v1: /root/autodl-tmp/AutoPlanner/results/shared/bufotalin_agentic_blackboard_full_retry_v7_20260609/r2_rank_analogical_hypotheses_analogical_hypothesis_ranking_v1.json`
- `r3_extract_visual_literature_chain_visual_literature_chain_extraction_result_v1: /root/autodl-tmp/AutoPlanner/results/shared/bufotalin_agentic_blackboard_full_retry_v7_20260609/r3_extract_visual_literature_chain_visual_literature_chain_extraction_result_v1.json`
- `r4_extract_visual_literature_chain_visual_literature_chain_extraction_result_v1: /root/autodl-tmp/AutoPlanner/results/shared/bufotalin_agentic_blackboard_full_retry_v7_20260609/r4_extract_visual_literature_chain_visual_literature_chain_extraction_result_v1.json`
- `r5_compile_exact_literature_rows_compiled_source_detail_chain_route_v1: /root/autodl-tmp/AutoPlanner/results/shared/bufotalin_agentic_blackboard_full_retry_v7_20260609/r5_compile_exact_literature_rows_compiled_source_detail_chain_route_v1.json`
- `r6_auto_failure_critic_failure_critic_report_v1: /root/autodl-tmp/AutoPlanner/results/shared/bufotalin_agentic_blackboard_full_retry_v7_20260609/r6_auto_failure_critic_failure_critic_report_v1.json`

Artifact bundle keys

- `agent_blackboard, analogical_hypothesis_ranking, compiled_downstream, compiled_downstream_payload, failure_critic_report, guided_chemenzy, literature_pdf_structure_evidence, literature_pdf_structure_evidence_dir, literature_scout, parent_route_proof, route_expansion_subgoal_search, route_failure_feedback, source_detail_chain_route, stitched_semisynthesis_route, visual_literature_chain_extraction, visual_structure_candidate_chain`